Modelling NOx formation in a lean premixed Methane-air combustor using a network of perfectly stirred reactors (PSRs) in Cantera

Faisal Zohad Sayyed

Colorado State University, Fort Collins, 80521, USA

In this study, a network of perfectly stirred reactors (PSRs) is used to numerically investigate the generation of nitrogen dioxide (NO_2) and nitric oxide (NO) in a lean premixed methane-air combustor. Cantera was used to model a five-stage PSR arrangement using the GRI-Mech 3.0 reaction mechanism [1]. The study assesses the effects of residence time (τ) and equivalency ratio (ϕ) on NOx emissions during combustion and post-flame gas evolution. The intricate relationship between NO and NO_2 and flow characteristics, thermal impacts, and chemical kinetics is captured by the simulation results. The study has useful implications for the design of gas turbine combustion systems [5] and describes an effective method for assessing emission profiles in a staged reactor setting.

1.Introduction

When building combustion systems for gas turbines, controlling NOx emissions is a significant problem, especially in lean-premixed situations where flame temperatures are maintained below stoichiometric values to limit NO production. Even though lean premixed combustion lowers peak temperatures, post-flame oxidation pathways can still produce a sizable amount of NO. Optimizing combustor performance and adhering to regulatory requirements require an understanding of how residence time and fuel-air ratio impact NOx output.

The Zel'dovich process, which is extremely sensitive to flame temperature, is the main method by which thermal NO is formed [4]. Conversely, slower secondary oxidation activities involving

NO and peroxy radicals are frequently the source of NO_2. Therefore, monitoring both NO and NO_2 is essential for a precise emissions analysis. A straightforward yet effective model for examining these effects is provided by perfectly stirred reactors (PSRs), which use discretized reaction zones to depict the course of combustion.

2. Methodology

2.1 Simulation Platform

In a staged reactor configuration, combustion chemistry was simulated using the open-source chemical kinetics toolkit Cantera[2]. The GRI-Mech 3.0 mechanism[1], which includes thorough routes for NOx, CO, hydrocarbon oxidation, and intermediate radicals, was implemented in all simulations. Jupyter Notebook with Anaconda was used to write and run the Python code [3].

2.2 Reactor Configuration

Five perfectly stirred reactors (PSRs) connected in series were used to represent the combustor. Every PSR in Cantera is an Ideal Gas Reactor, which models a well-mixed gas system with a constant volume. Each reactor's output composition is used as the input for the subsequent one.

2.3 Operating Conditions

- Fuel: Methane (CH_4)
- Oxidizer: Air represented by O_2: N_2 = 1: 3.76
- Initial Temperature (T_0): 800 K
- Pressure (P_0): 20 bar
- Equivalence Ratios (ϕ): [0.6, 0.7, 0.8, 0.9, 1.0] [6]
- Residence Times (τ): [0.002 s, 0.005 s, 0.01 s, 0.02 s]

2.4 Equivalence Ratio Calculation

The equivalence ratio (ϕ) is given by:

$\phi = F/A \div (F/A)_{stoi} c_h$

The ratio of the actual fuel-to-air ratio to the stoichiometric fuel-to-air ratio is known as the equivalency ratio (ϕ)[6].

The following command was used in Cantera to set the composition of the mixture:

Python--

```
gas.set_equivalence_ratio(phi, "CH4", {"O2": 1, "N2": 3.76})
```

2.5 Residence Time Modeling

The Cantera ReactorNet advance method was used to increase the simulation time by the residence time (τ) for each reactor stage:

Python—

```
sim.advance(sim.time + tau)
```

By holding the gas for a predetermined amount of time before moving it on to the next step, each perfectly stirred reactor (PSR) replicates the gradual passage of gases through combustion zones.

2.6 Output Variables

- NO and NO_2 mole fractions were taken out of each reactor stage.
- All possible combinations of residence time (τ) and equivalency ratio (ϕ) were simulated
- Using Python's Matplotlib module, output data was saved and displayed.

3. Results

Different patterns of NO and NO_2 production throughout the PSR network were identified by the simulation:

- **NO:** As the reactor progressed, concentrations gradually elevated. As the combination became closer to stoichiometric conditions, the NO mole fractions started to slightly decrease after rising around intermediate equivalence ratios (ϕ = 0.7–0.8).
- **NO_2:** Especially in later reactor stages, it showed a monotonic increase with ϕ. This suggests that the availability of NO and reactive radical species in the post-flame environment is what drives the creation of NO_2.

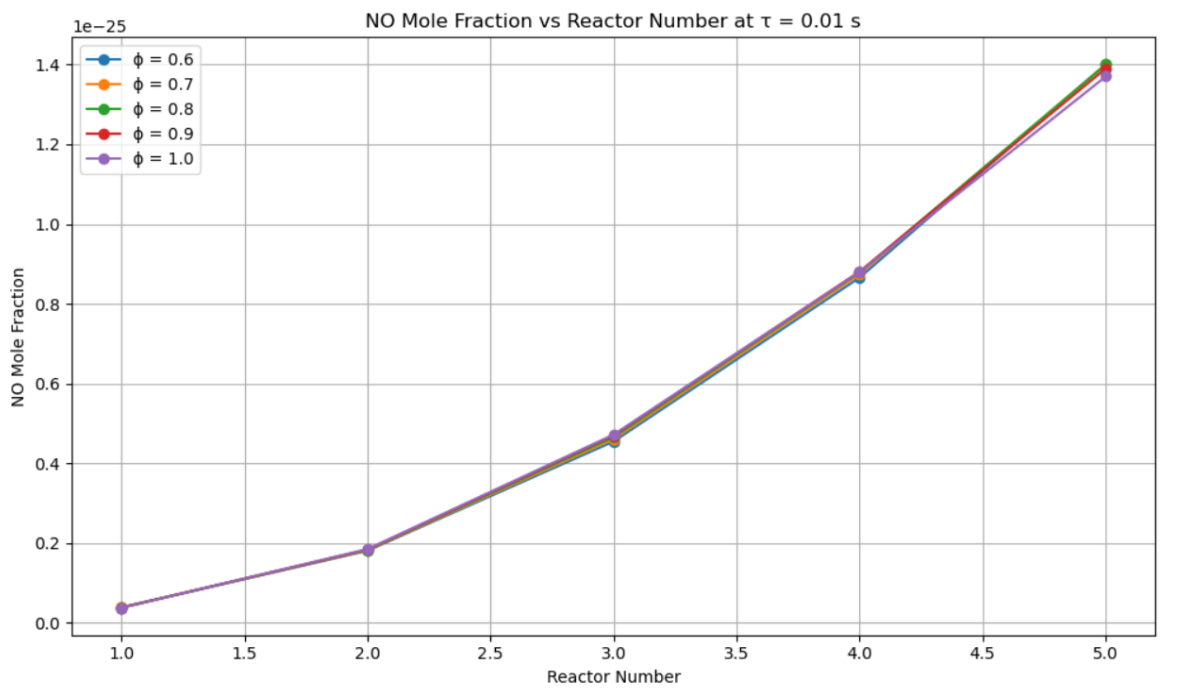

Fig 1: *Evolution of the NO mole fraction at a fixed residence period of 10 ms along a 5-stage PSR network. Different equivalency ratios (ϕ) are represented by each curve. The consistent increase across reactors suggests that post-flame temperature and oxygen availability are driving the gradual production of NO.*

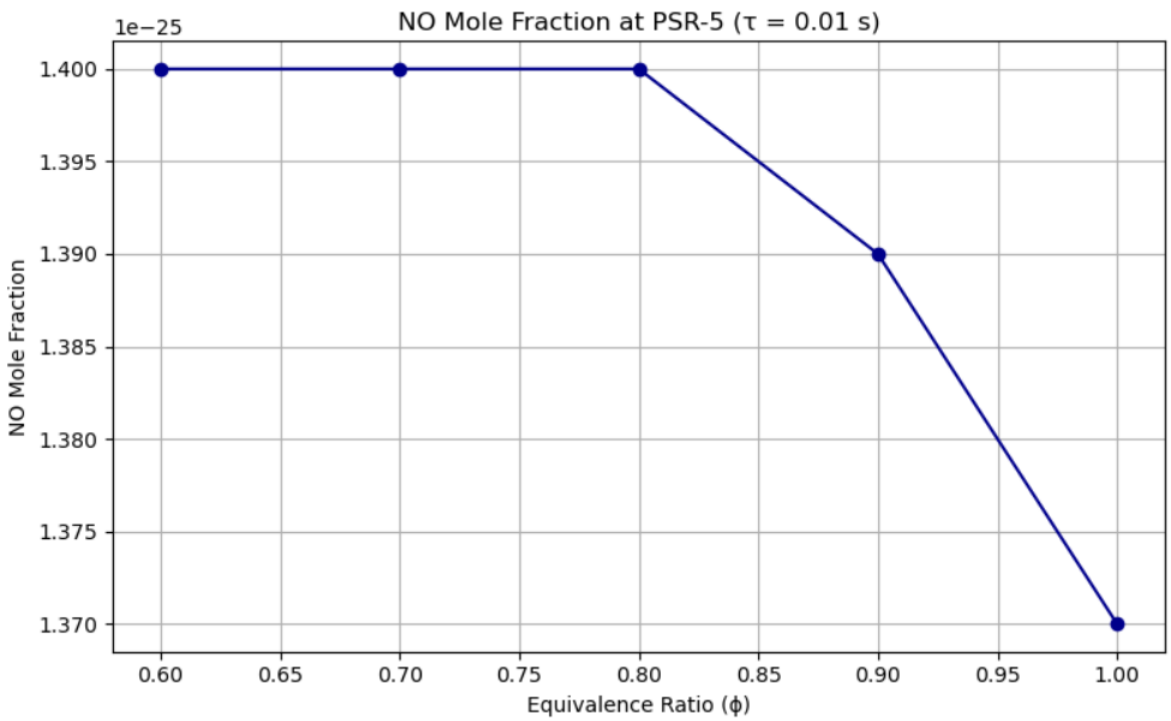

Fig 2: *NO mole fraction variation at the last PSR (Reactor 5) for τ = 10 ms as a function of equivalency ratio (ϕ). The graph shows that NO production is comparatively constant in the lean regime (ϕ = 0.6–0.8), with a minor decline as one approaches stoichiometric circumstances because post-flame NO chemistry has a shorter residence period.*

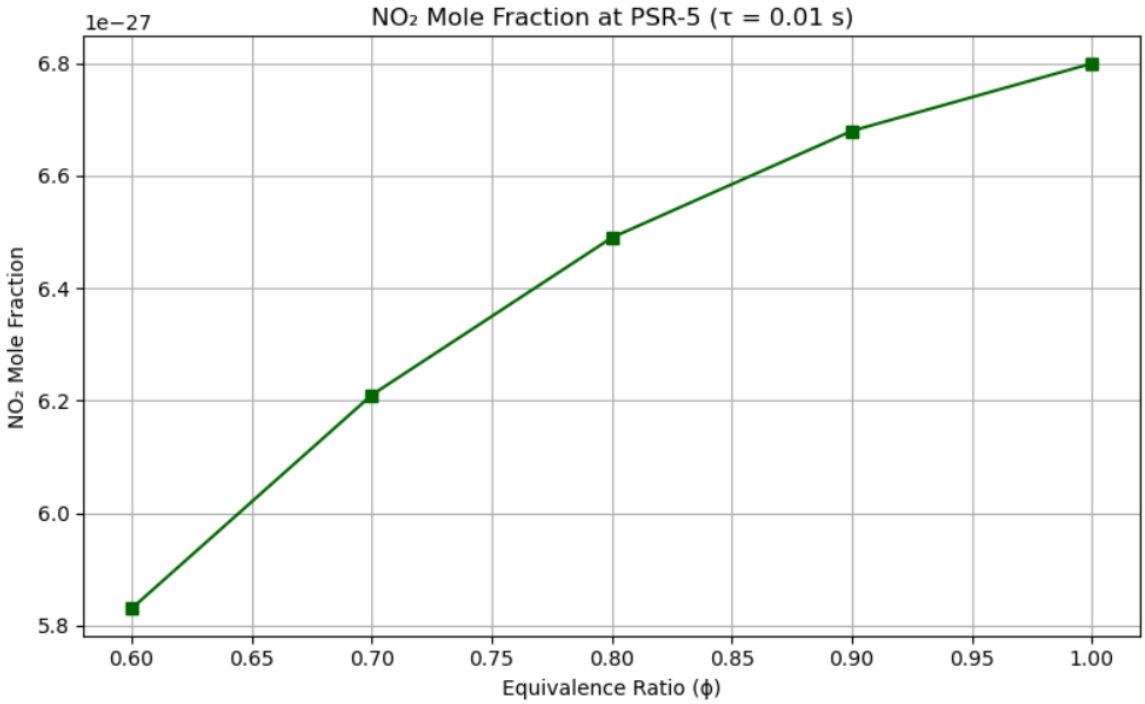

Fig 3: *NO_2 mole fraction for different equivalency ratios at τ = 10 ms at the PSR network's (Reactor 5) outlet. Because of more active post-flame chemistry and increased NO availability, richer mixes exhibit improved NO_2 production, as evidenced by the increasing trend.*

[2]:

	phi	tau (s)	Reactor	NO	NO2
10	0.6	0.01	1	3.90e-27	8.52e-30
11	0.6	0.01	2	1.81e-26	1.44e-28
12	0.6	0.01	3	4.55e-26	7.56e-28
13	0.6	0.01	4	8.66e-26	2.42e-27
14	0.6	0.01	5	1.40e-25	5.83e-27
30	0.7	0.01	1	3.87e-27	9.44e-30
31	0.7	0.01	2	1.82e-26	1.59e-28
32	0.7	0.01	3	4.61e-26	8.30e-28
33	0.7	0.01	4	8.74e-26	2.62e-27
34	0.7	0.01	5	1.40e-25	6.21e-27
50	0.8	0.01	1	3.83e-27	1.03e-29
51	0.8	0.01	2	1.83e-26	1.72e-28
52	0.8	0.01	3	4.66e-26	8.94e-28
53	0.8	0.01	4	8.79e-26	2.78e-27
54	0.8	0.01	5	1.40e-25	6.49e-27

Table 1: *At a residence duration of 10 ms, the NO and NO_2 mole fractions were projected for five PSRs with different equivalency ratios. The findings indicate that NOx generation increases with increasing φ, which is in line with thermal NOx processes in lean premixed circumstances.*

4.Code Implementation

Using Cantera, the basic simulation was run in Python and looped over a variety of residency periods and equivalency ratios. The PSR configuration and NOx extraction logic are described in the script below:

jupyter Untitled Last Checkpoint: 8 minutes ago

File Edit View Run Kernel Settings Help

Trusted

Code

JupyterLab Python 3 (ipykernel)

[1]:
```python
import cantera as ct
import numpy as np
import matplotlib.pyplot as plt

# Simulation parameters
phi_values = [0.6, 0.7, 0.8, 0.9, 1.0]
residence_times = [0.002, 0.005, 0.01, 0.02]
n_reactors = 5

# Setup gas object using GRI-Mech 3.0
gas = ct.Solution("gri30.yaml")
fuel = "CH4"
oxidizer = {"O2": 1.0, "N2": 3.76}
T0 = 800  # Initial temperature in Kelvin
P0 = 2e6  # Pressure in Pa (20 bar)

results = []

for phi in phi_values:
    for tau in residence_times:
        gas.set_equivalence_ratio(phi, fuel, oxidizer)
        gas.TP = T0, P0

        reactors = [ct.IdealGasReactor(gas) for _ in range(n_reactors)]
        sim = ct.ReactorNet(reactors)

        for i in range(n_reactors):
            sim.advance(sim.time + tau)
            no = reactors[i].thermo["NO"].X[0]
            no2 = reactors[i].thermo["NO2"].X[0]
            results.append((phi, tau, i+1, no, no2))

# Output: phi, tau, reactor number, NO, NO2 mole fractions
for row in results:
    print(f"φ={row[0]:.2f}, τ={row[1]:.3f}s, PSR-{row[2]}: NO={row[3]:.2e}, NO2={row[4]:.2e}")
```

5. Discussion

The outcomes are consistent with the Zel'dovich thermal NO mechanism's predicted behavior:

1. $O + N_2 \rightarrow NO + N$

2. $N + O_2 \rightarrow NO + O$

3. $N + OH \rightarrow NO + H$

Peak flame temperature decreases at lean equivalency ratios, which causes NO production to occur more slowly. On the other hand, residence time increases NO accumulation in reactors downstream. The oxidation of NO results in the more slow production of NO_2:

$NO + HO_2 \rightarrow NO_2 + OH$

Both NO and NO_2 dramatically rise with increasing residence duration, underscoring the need of flow residence control in combustor design. Additionally, trends indicate that even slight changes

in ϕ can have a big effect on emission output, which emphasizes the necessity of careful equivalency ratio control.

6.Conclusion

This paper presents a computational approach that uses a multi-stage PSR network to describe NOx production in a lean premixed methane-air combustor. The approach effectively captures the nonlinear behavior of NO and NO_2 emissions under conditions relevant to gas turbines by resolving chemical evolution over five interconnected reactor stages and altering operational parameters like equivalency ratio (ϕ) and residence duration (τ).

Important findings show that nitric oxide (NO) generation rises steadily across the reactor sequence, peaking close to equivalency ratios of $\phi = 0.7–0.8$. Because of changed reaction routes and decreased oxygen availability, NO concentrations plateau or fall at greater equivalency ratios. On the other hand, subsequent oxidation of NO in the post-flame environment is the main cause of the monotonic increase in nitrogen dioxide (NO_2) with increasing ϕ. A longer τ allows for a longer reaction time and a greater concentration of NOx species downstream, making the effect of residence time more important.

This PSR-based method maintains comprehensive kinetic integrity while offering a computationally effective substitute for full-scale CFD simulations. It successfully separates the impact of reactor staging and chemical timeframes on the production of pollutants. The findings lend credibility to the wider use of staged reactor models in emission control plans and pre-screening combustor layouts[5].

For better predictive capabilities in real-world combustor designs, further research may expand this methodology by adding spatial temperature gradients, pressure sensitivity analysis, and coupling with transport-resolved CFD solvers[7].

7.references

[1] Smith, G. P., et al. GRI-Mech 3.0 Mechanism. http://www.me.berkeley.edu/gri_mech/

[2] Kee, R. J., et al. Chemically Reacting Flow: Theory and Practice, Wiley, 2003.

[3] Cantera Documentation. Cantera/Anaconda

[4] Turns, S. R., An Introduction to Combustion: Concepts and Applications, 3rd ed., McGraw-Hill, 2012.

[5] Lefebvre, A. H., and Ballal, D. R., Gas Turbine Combustion: Alternative Fuels and Emissions, 3rd ed., CRC Press, 2010.

[6] Warnatz, J., Maas, U., and Dibble, R. W., Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, 4th ed., Springer, 2006.

[7] Ihme, M., “Combustion and Engine Core-Noise,” Annual Review of Fluid Mechanics, vol. 45, pp. 437–467, 2013.